

ECEN 474/704: Lab 01

Introduction to Cadence and MOS Device Characterization

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1 I. DISCUSSION OF THE LAB AND RESULTS

In this lab, the transistor characterization experiments were performed for both NMOS and PMOS Devices under different width to length ratios. There are 3 subdivisions for both nmos and pmos 'a' deals with sweeping gate voltages and obtaining the id current, 'b' deals with sweeping Vds and plotting id and related graph work, this part involves a half dc and a half ac analysis, where unity gain frequency is calculated and section c is purely AC where a current source is connected to the drain terminal and frequency analysis is done and the intrinsic gain is calculated.

II. SUMMARY OF RESULTS

Summarizing the simulation results from the lab in one or two tables, like what is shown below.

Table 1

Example summary of results for process parameter characterization for nfetx

Parameter	$\mu C_{ox} \mu A / v^2$	$V_{th}(mV)$	θ	$\lambda(u v^{(-1)})$	$f_T(GHz)$	$A_i(db)$
W/L=0.5 μ /0.18 μ using 1 finger.	167.292	460.433	0.9235	5.906	46.2306	27
W/L=5 μ /0.18 μ using 1 finger.	217.188	459.995	1.294	54.49	43.3905	27
W/L 5 μ /0.18 μ using 10 fingers	181.188	442.293	0.93612	62.639	49.4994	27.155
W/L=5 μ /0.27 μ	214.326	447.0368	0.9942	20.997	34.1123	24.2463

Table 1

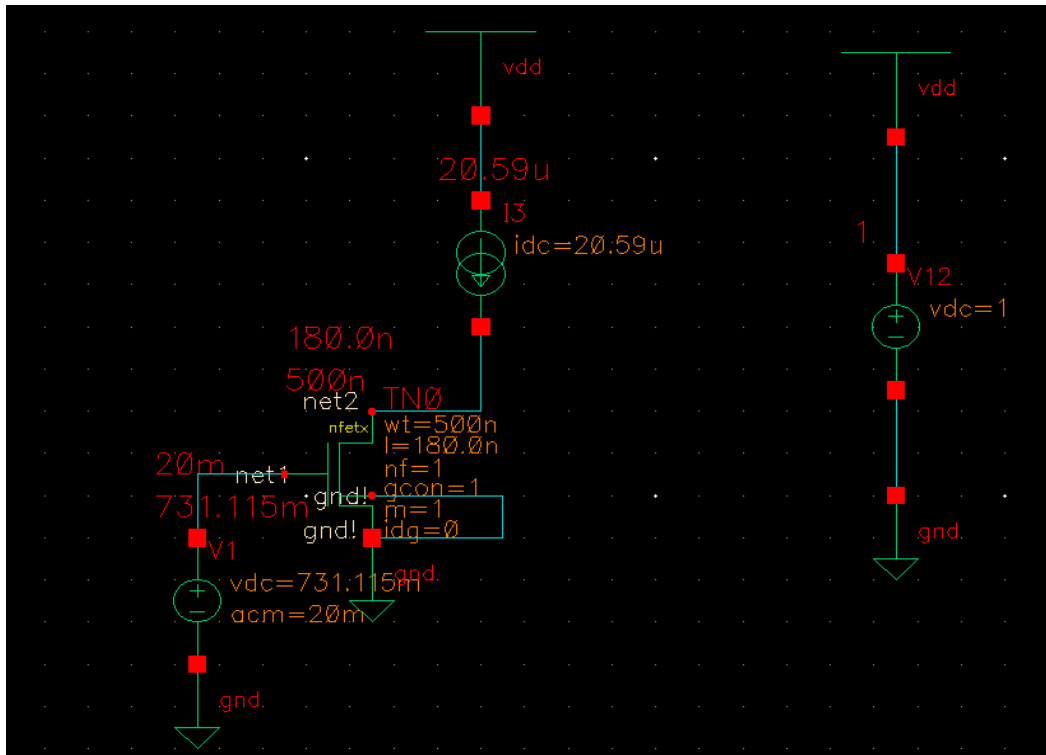
Example summary of results for process parameter characterization for pfetx

Parameter	$\mu C_{ox} \mu A / v^2$	$V_{th}(mV)$	θ	$\lambda(u v^{(-1)})$	$f_T(GHz)$	$A_i(db)$
W/L=0.5 μ /0.18 μ using 1 finger	-47.376	432.271	0.7483	-2.6	14.9343	-10.36625
W/L=5 μ /0.18 μ using 1 finger.	-56.48	414.2959	0.7744	-29.82	16.6254	-10.55418
W/L 5 μ /0.18 μ using 10 fingers	-47.376	432.3074	0.7485	-26.2935	14.9351	-10.36581
W/L=5 μ /0.27 μ using 1 finger	-47.52	421.6376	0.65524	-11.873	7.55219	-12

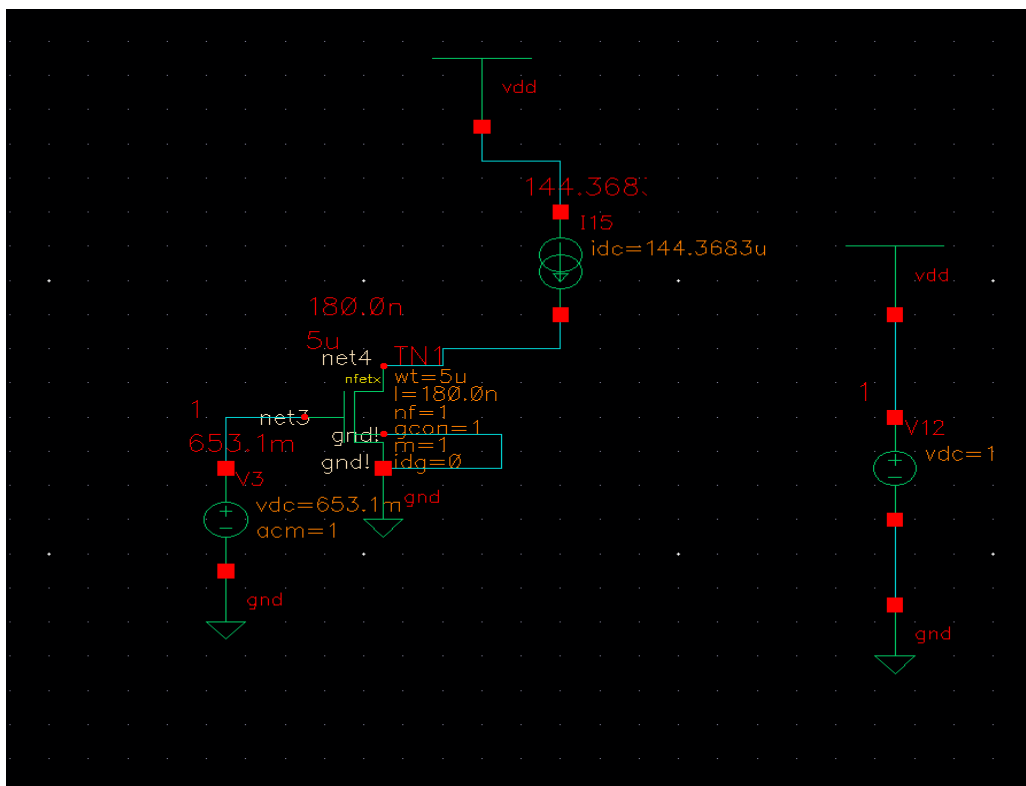
III. SUB-CIRCUIT SCHEMATICS

1. NMOS(nfetx):

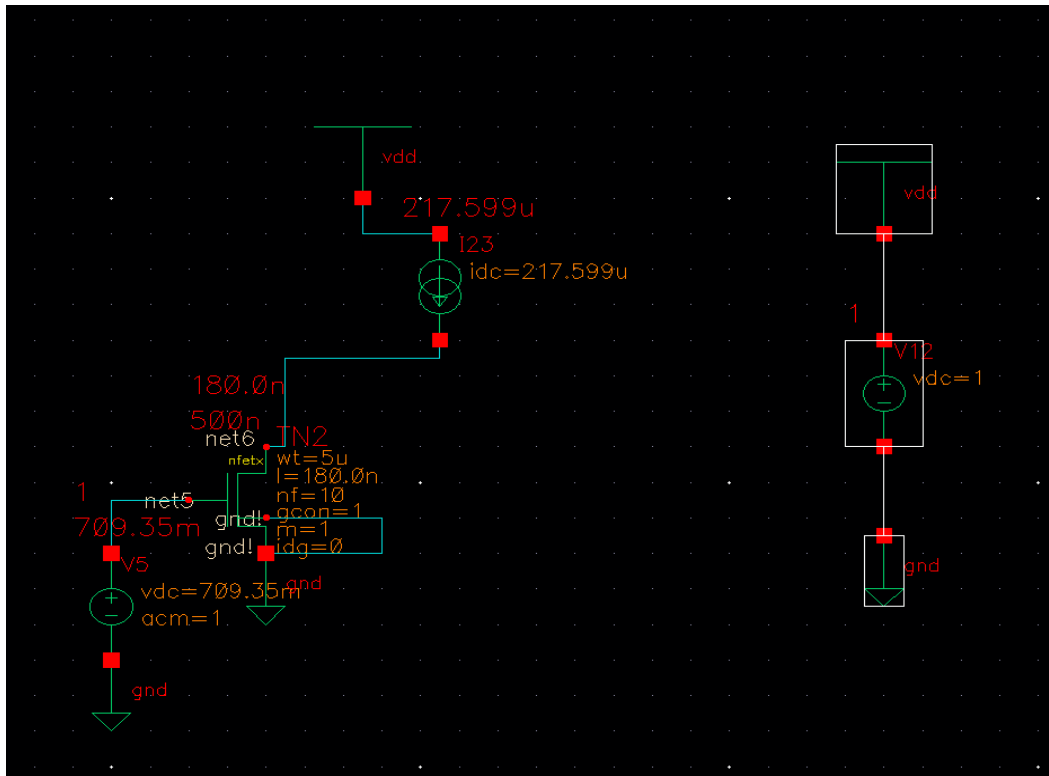
1.1. W/L=0.5 μ /0.18 μ using 1 finger:



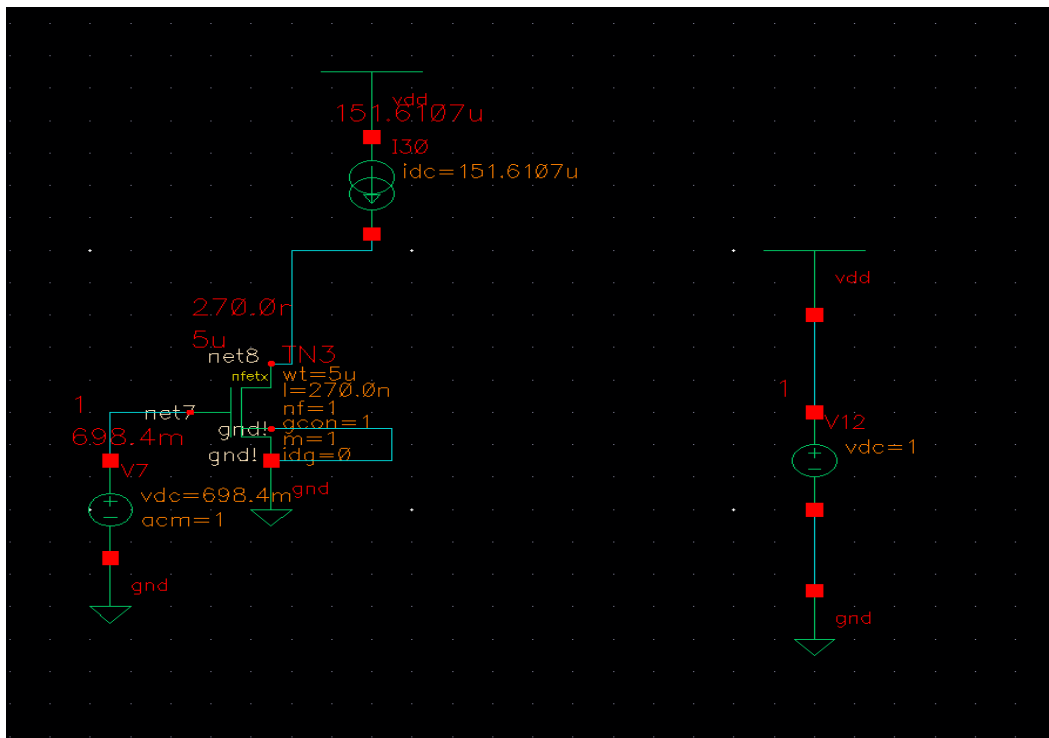
1.2. $W/L = 5\mu/0.18\mu$ using 1 finger.



1.3. $W/L = 5\mu/0.18\mu$ using 10 fingers

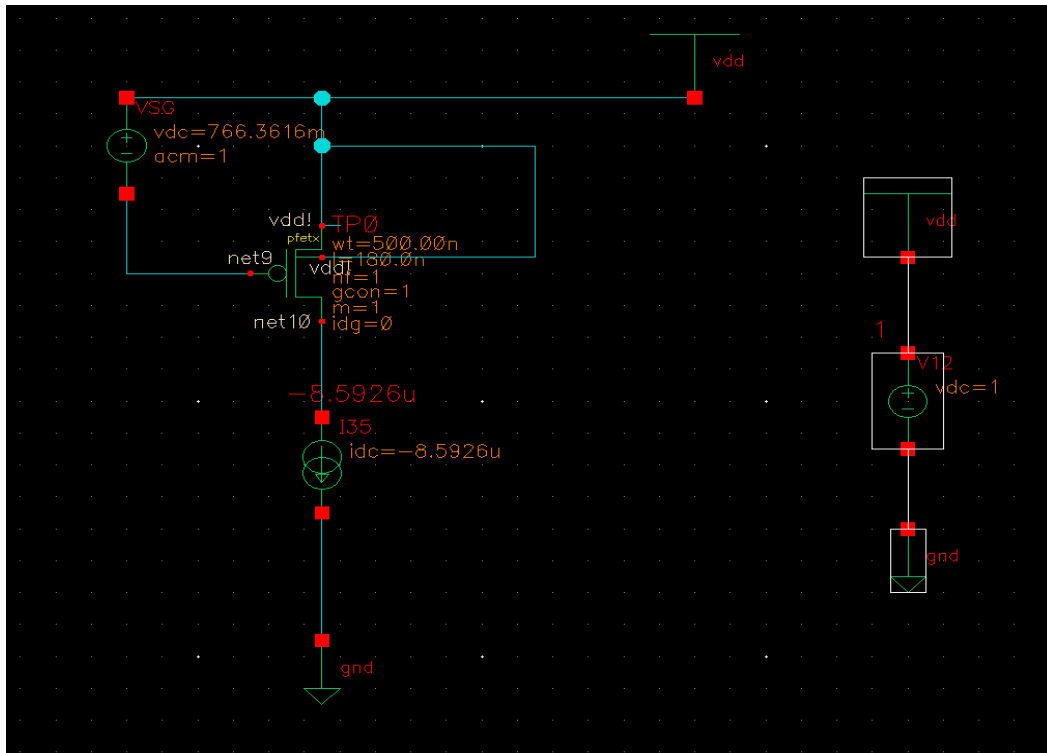


1.4. $W/L=5\mu/0.27\mu$ using 1 finger.

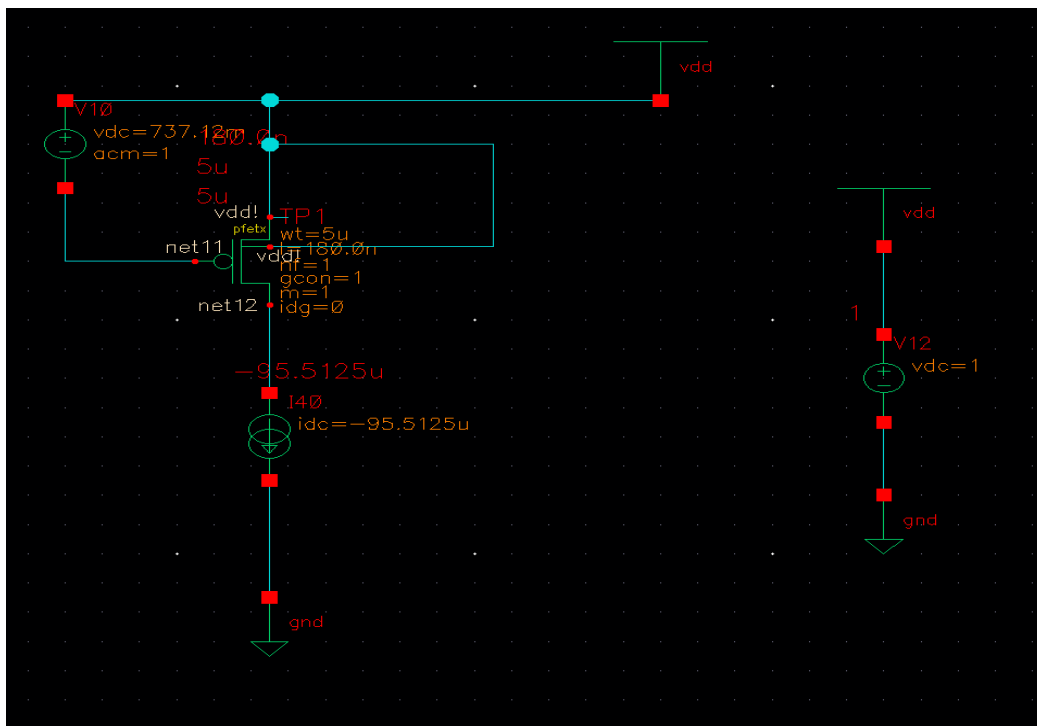


2. PMOS(pfetx):

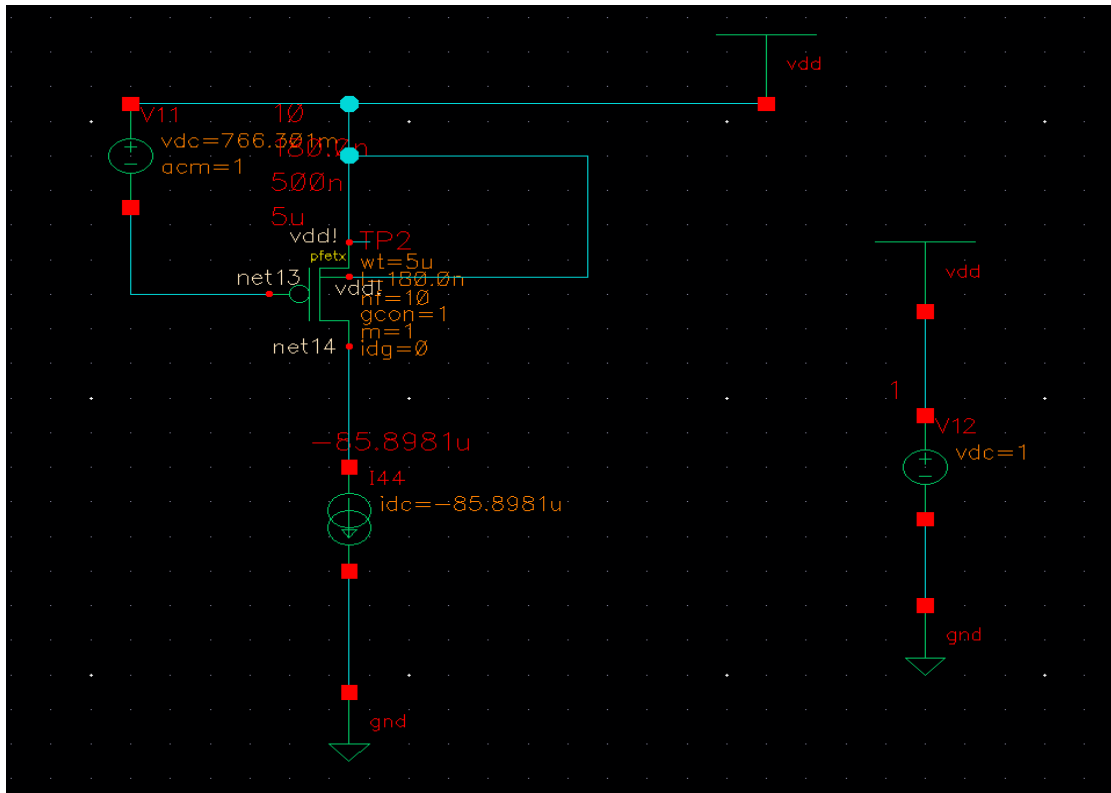
2.1. $W/L=0.5\mu/0.18\mu$ using 1 finger



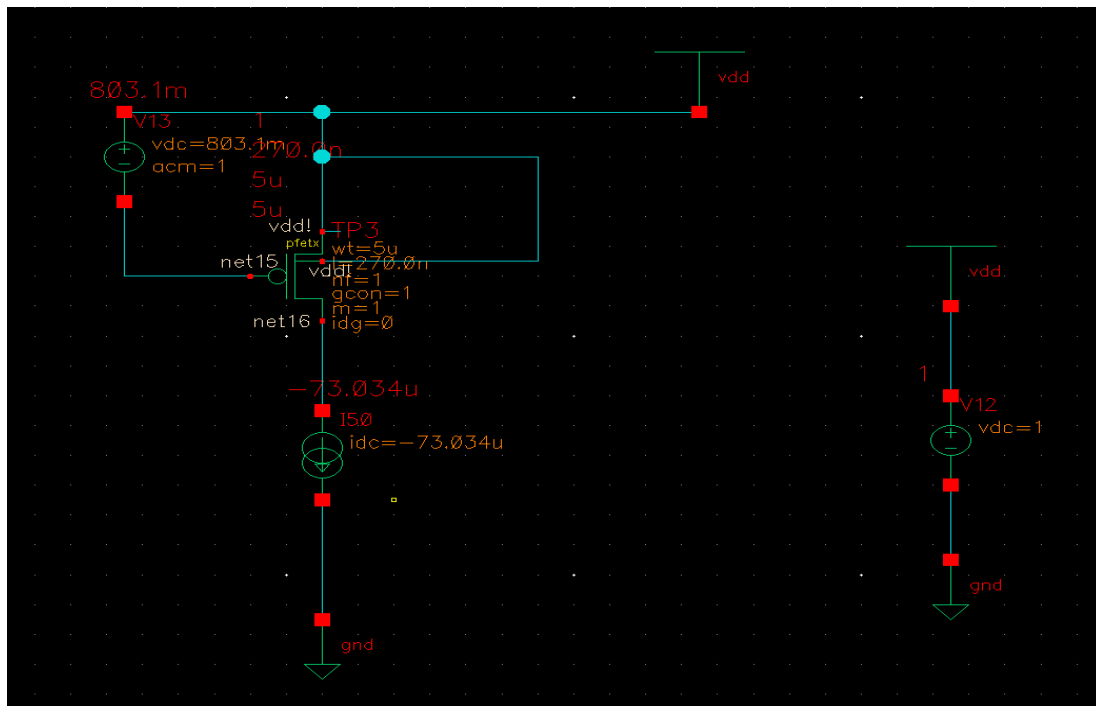
2.2. $W/L=5\mu/0.18\mu$ using 1 finger.



2.3. $W/L=5\mu/0.18\mu$ using 10 fingers



2.4. $W/L = 5\mu/0.27\mu$ using 1 finger.

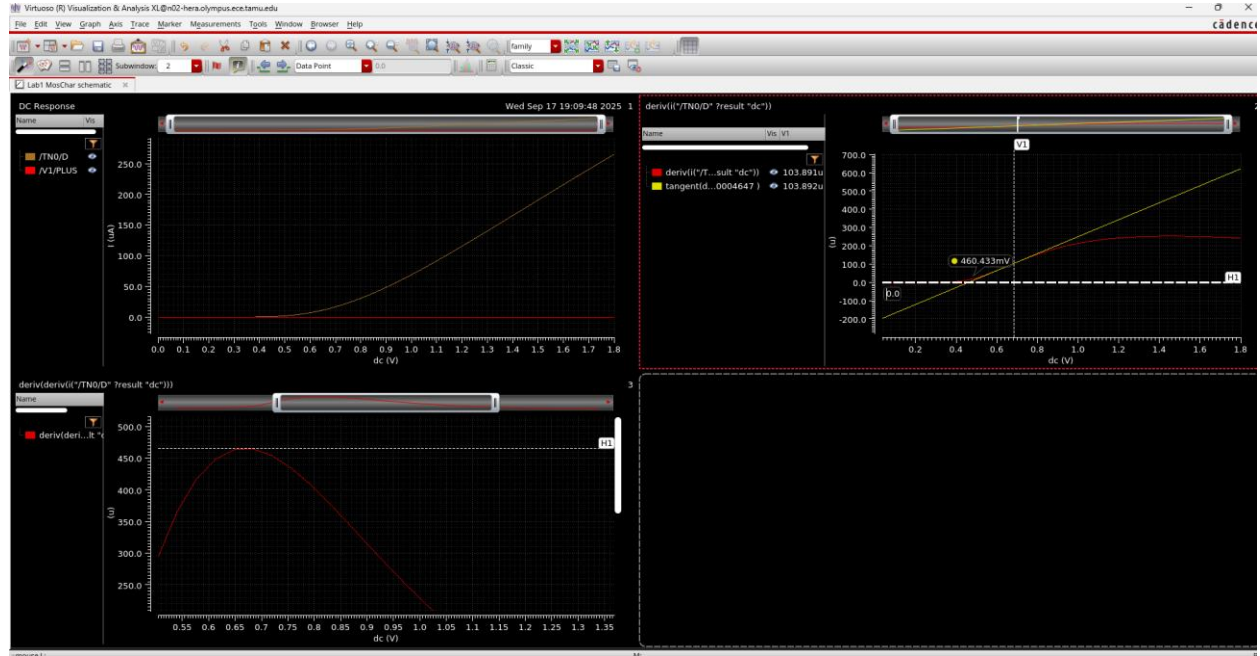


IV. SIMULATION RESULTS

1NMOS(nfetx):

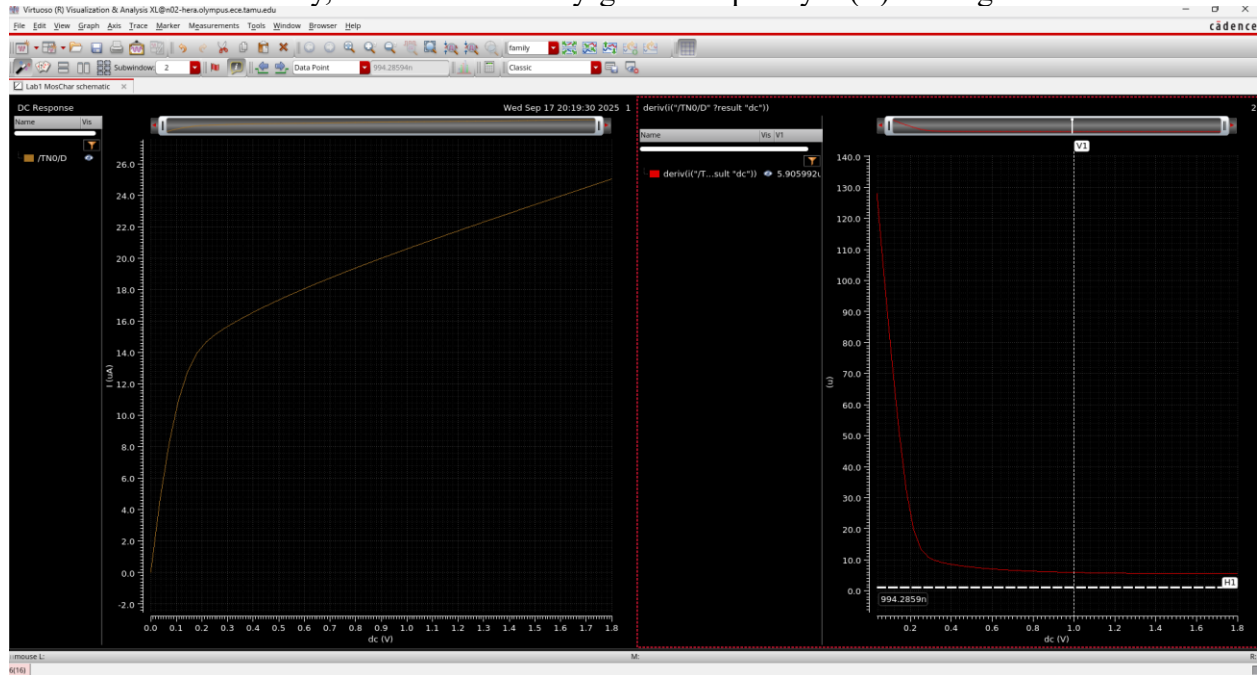
1. $W/L=0.5\mu/0.18\mu$ using 1 finger:

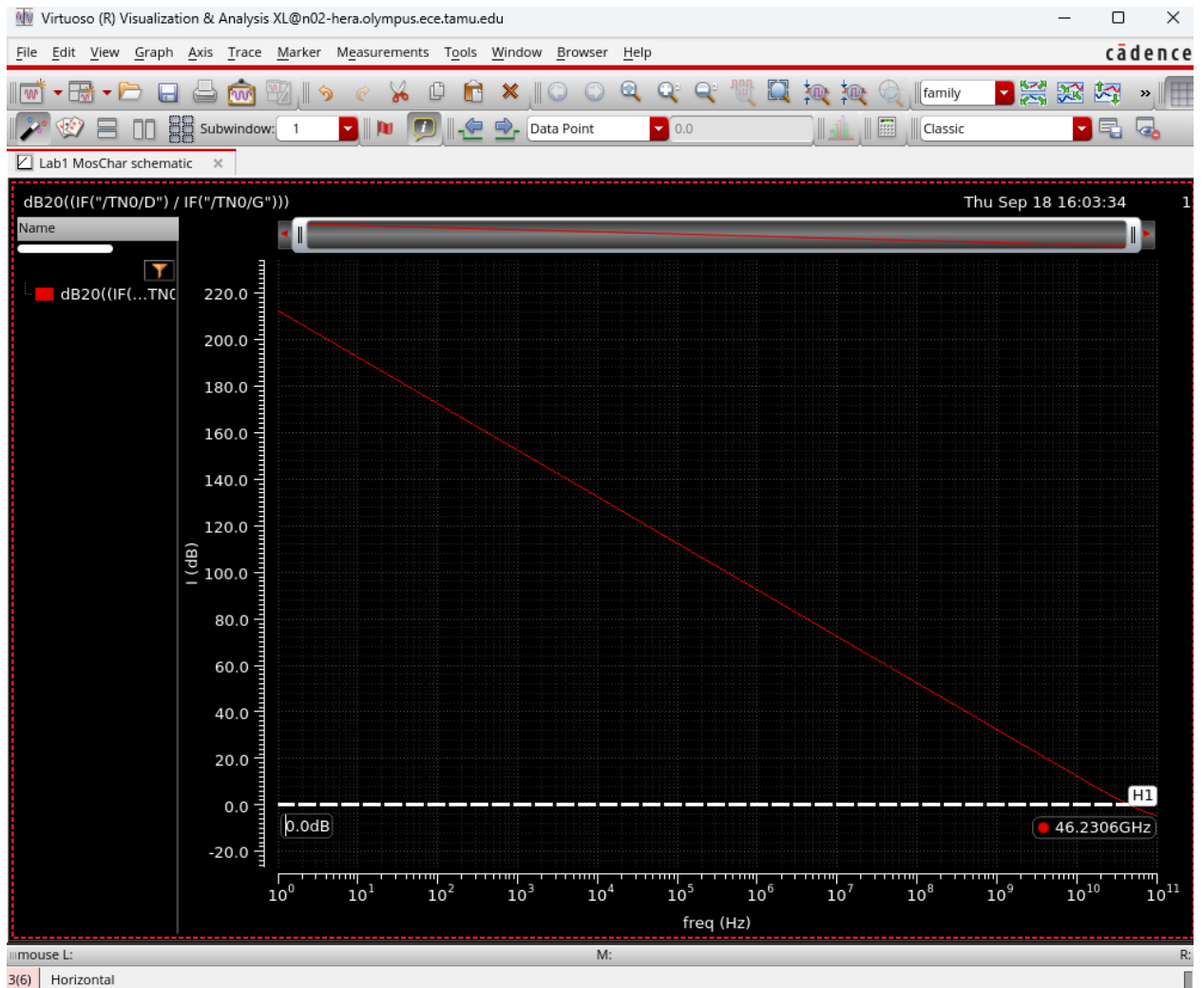
a. $V_{DS}=1V$. Sweep V_{GS} from 0 to 1.8V, plot (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .



b.

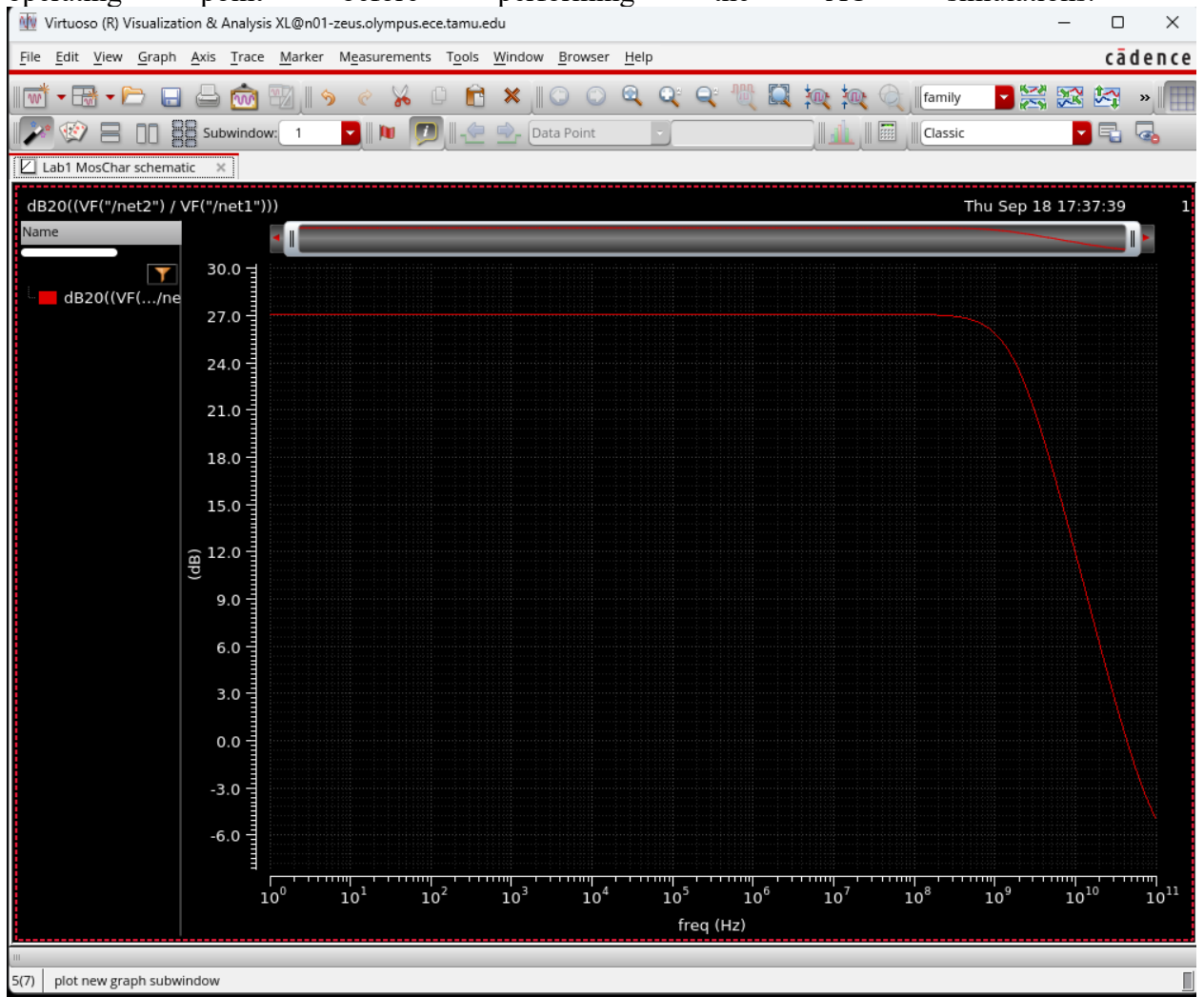
$G_S=V_{tn}+1/(40)$. Sweep V_{DS} from 0 to 1.8V, plot $(dI_D/dV_{DS})/I_D$ vs. V_{DS} and extract λ at $V_{DS}=1V$. Additionally, find the unity-gain frequency (f_t) using AC simulations.





c. Reconfigure the circuit if necessary and find the intrinsic gain (A_i) using AC simulations when $V_{DS}=1\text{V}$ and $V_{GS}=V_{tn}+1/(4\theta)$. Refer to Appendix A for a method of checking the DC

operating point before performing the AC simulations.

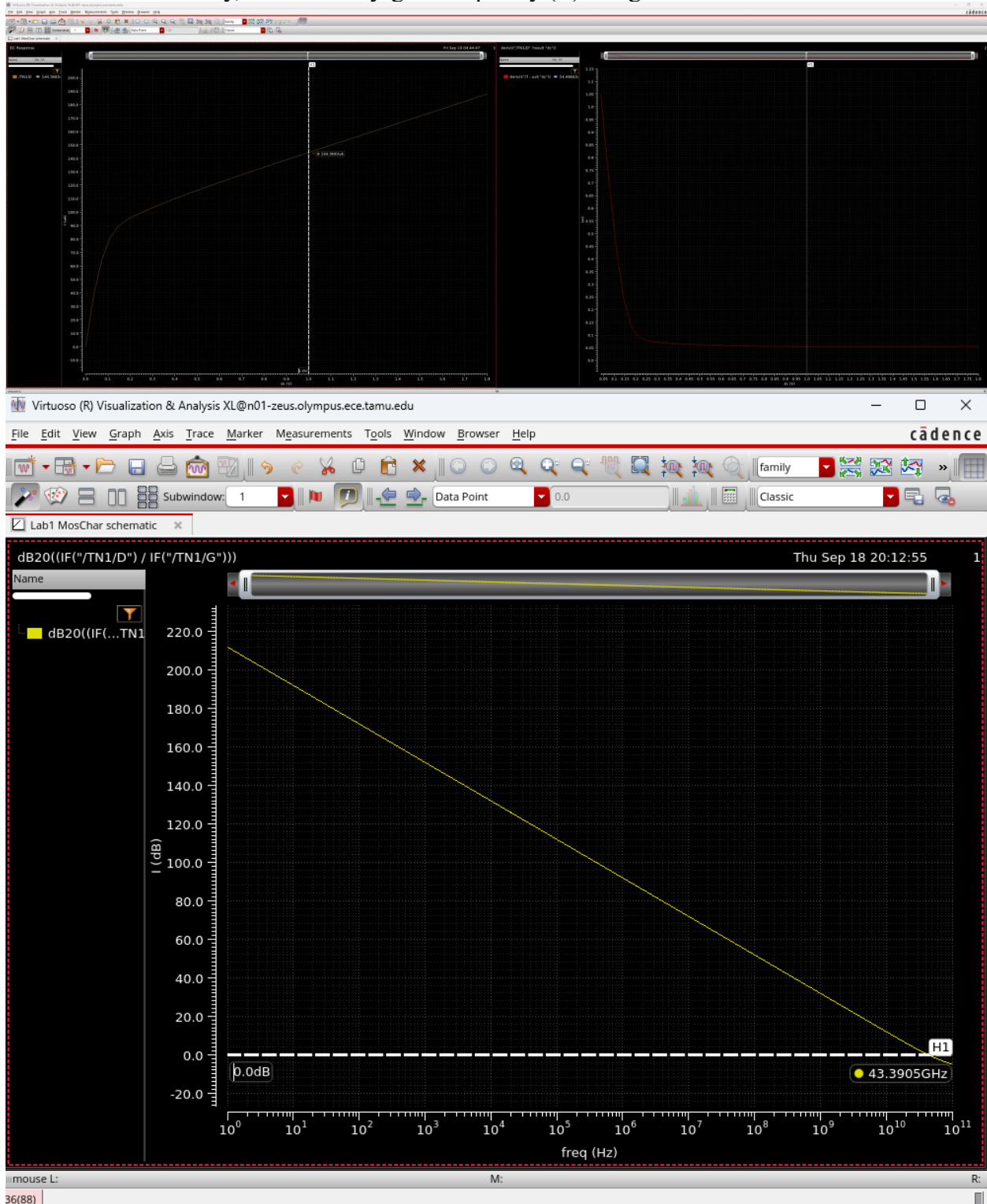


2. $W/L=5\mu/0.18\mu$ using 1 finger.

a. $V_{DS}=1V$. Sweep V_{GS} from 0 to 1.8V, plot (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ

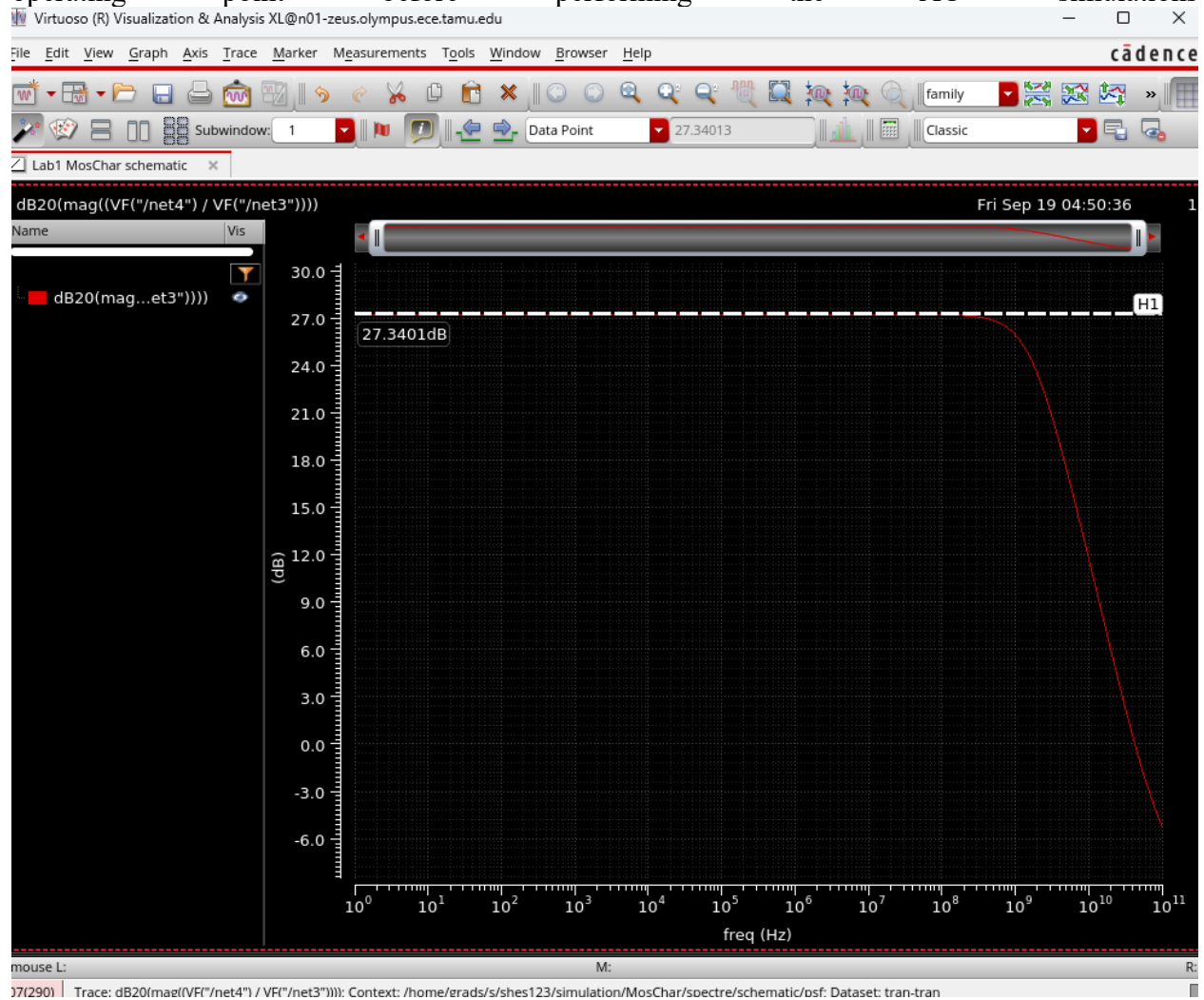


b. $V_{GS} = V_{tn} + 1/(40)$. Sweep V_{DS} from 0 to 1.8V, plot $(dI_D/dV_{DS})/I_D$ vs. V_{DS} and extract λ at $V_{DS}=1V$. Additionally, find the unity-gain frequency (f_t) using AC simulations.



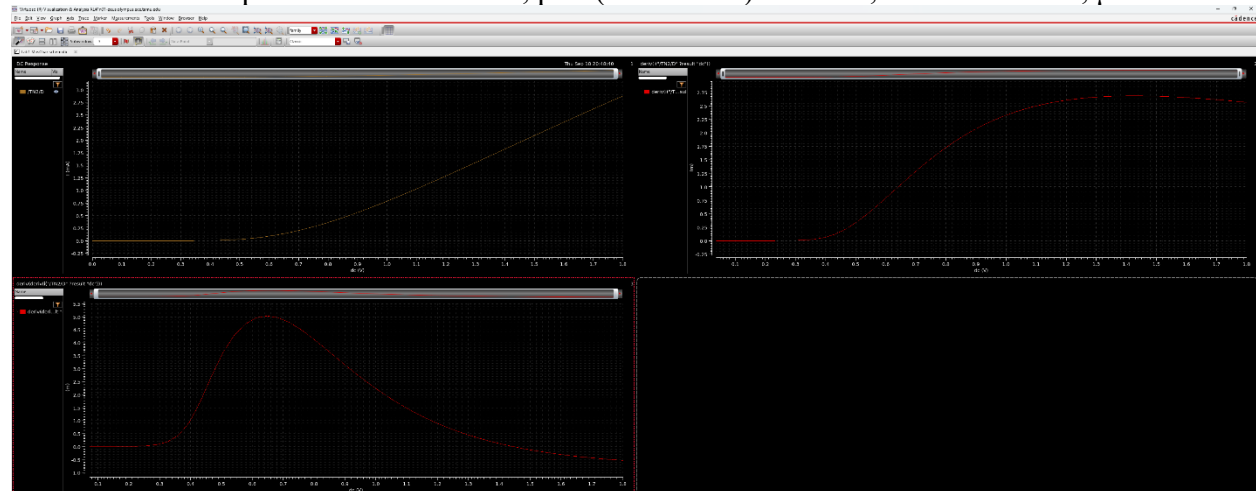
C. Reconfigure the circuit if necessary and find the intrinsic gain (A_i) using AC simulations when $V_{DS}=1V$ and $V_{GS}=V_{tn}+1/(40)$. Refer to Appendix A for a method of checking the DC

operating point before performing the AC simulations

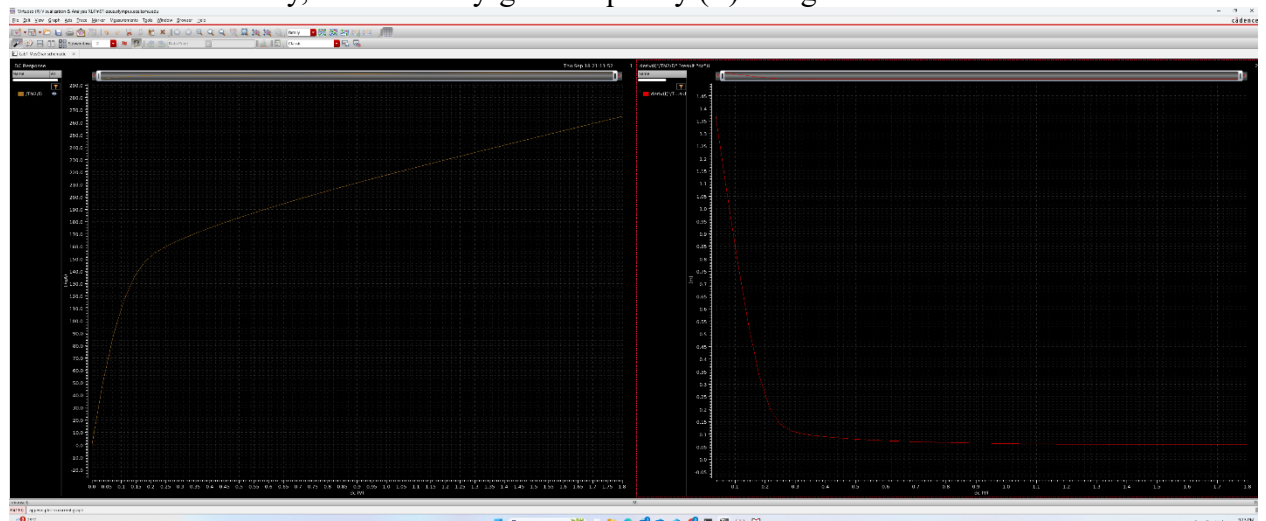


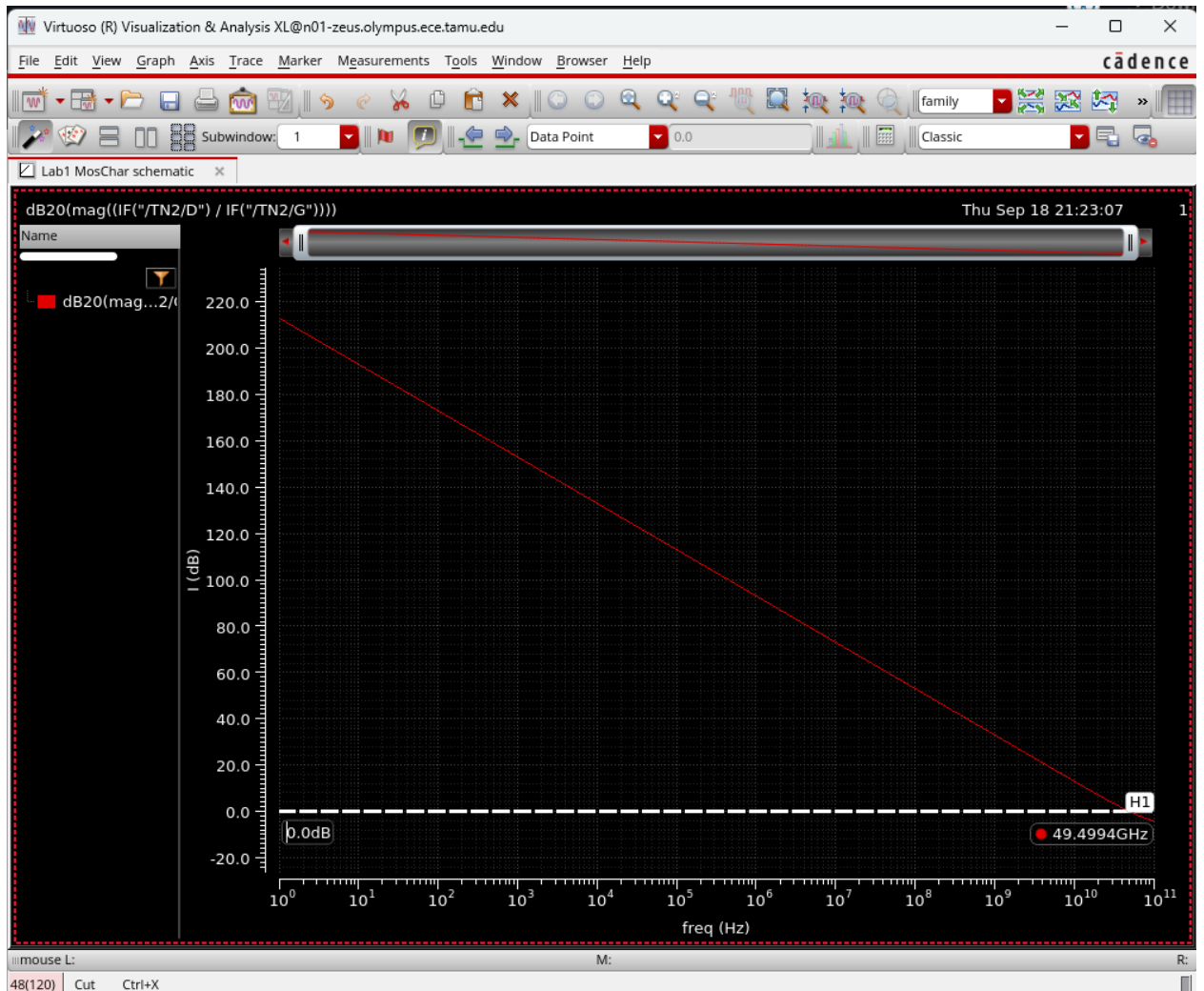
3. W/L $5\mu/0.18\mu$ using 10 fingers

a. VDS=1V. Sweep VGS from 0 to 1.8V, plot (dID/dVGS) vs. VGS, and extract V_{tn} , $\mu_n C_{ox}$ and θ

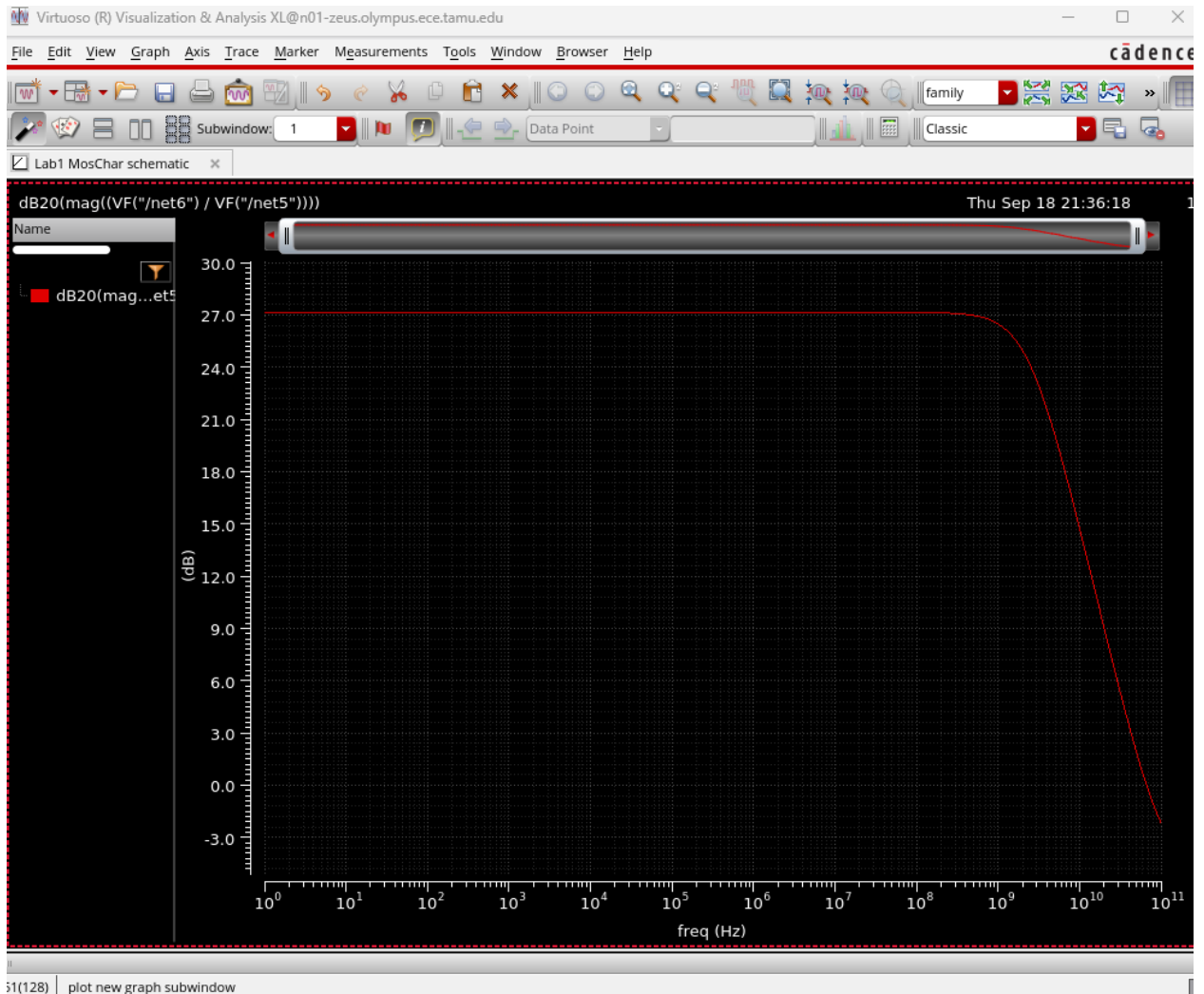


b. $V_{GS} = V_{tn} + 1/(40)$. Sweep V_{DS} from 0 to 1.8V, plot $(dI_D/dV_{DS})/I_D$ vs. V_{DS} and extract λ at $V_{DS} = 1V$. Additionally, find the unity-gain frequency (f_t) using AC simulations.



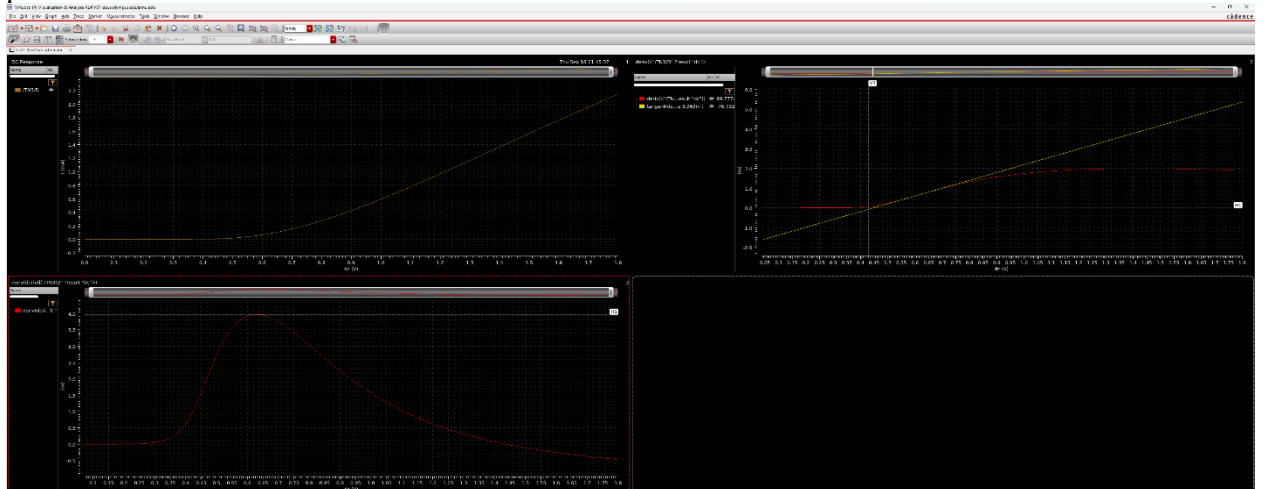


c. Reconfigure the circuit if necessary and find the intrinsic gain (A_i) using AC simulations when $V_{DS}=1\text{V}$ and $V_{GS}=V_{tn}+1/(4\theta)$. Refer to Appendix A for a method of checking the DC operating point before performing the AC simulations

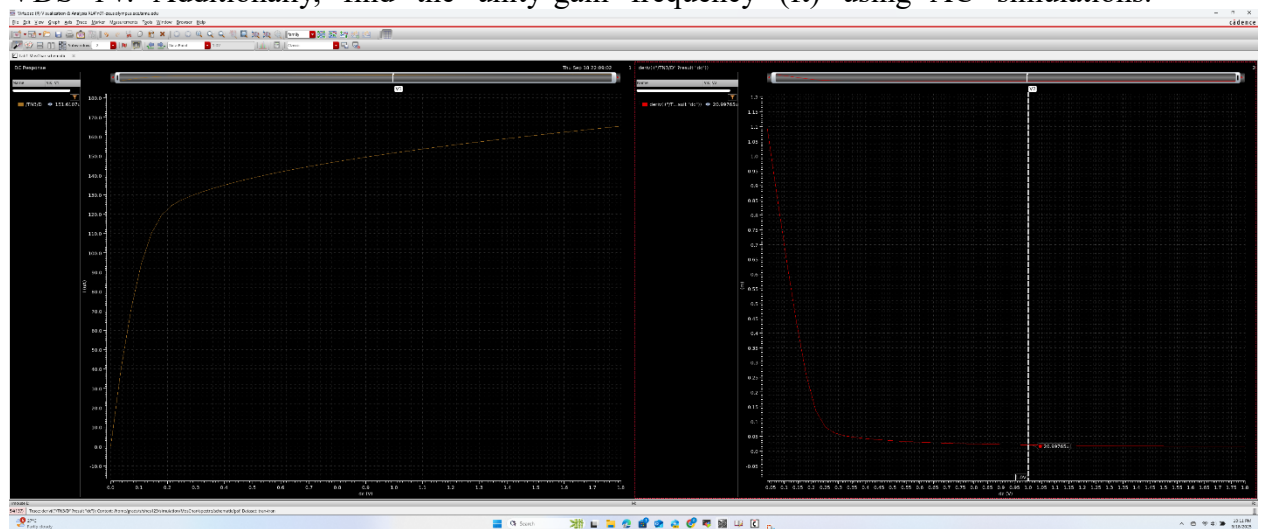


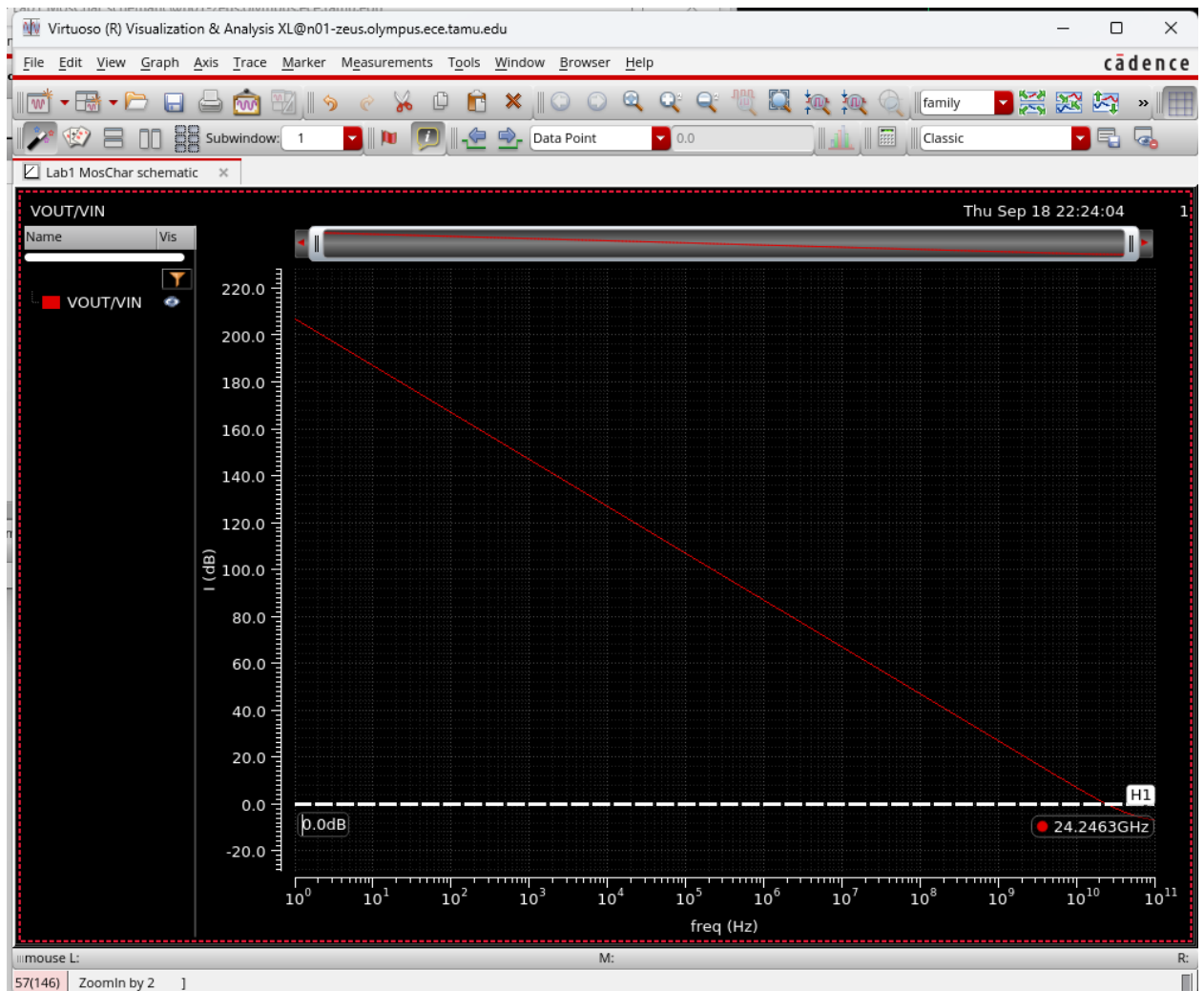
4. $W/L = 5\mu/0.27\mu$ using 1 finger.

a. $V_{DS} = 1V$. Sweep V_{GS} from 0 to 1.8V, plot (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , μnC_{ox} and θ

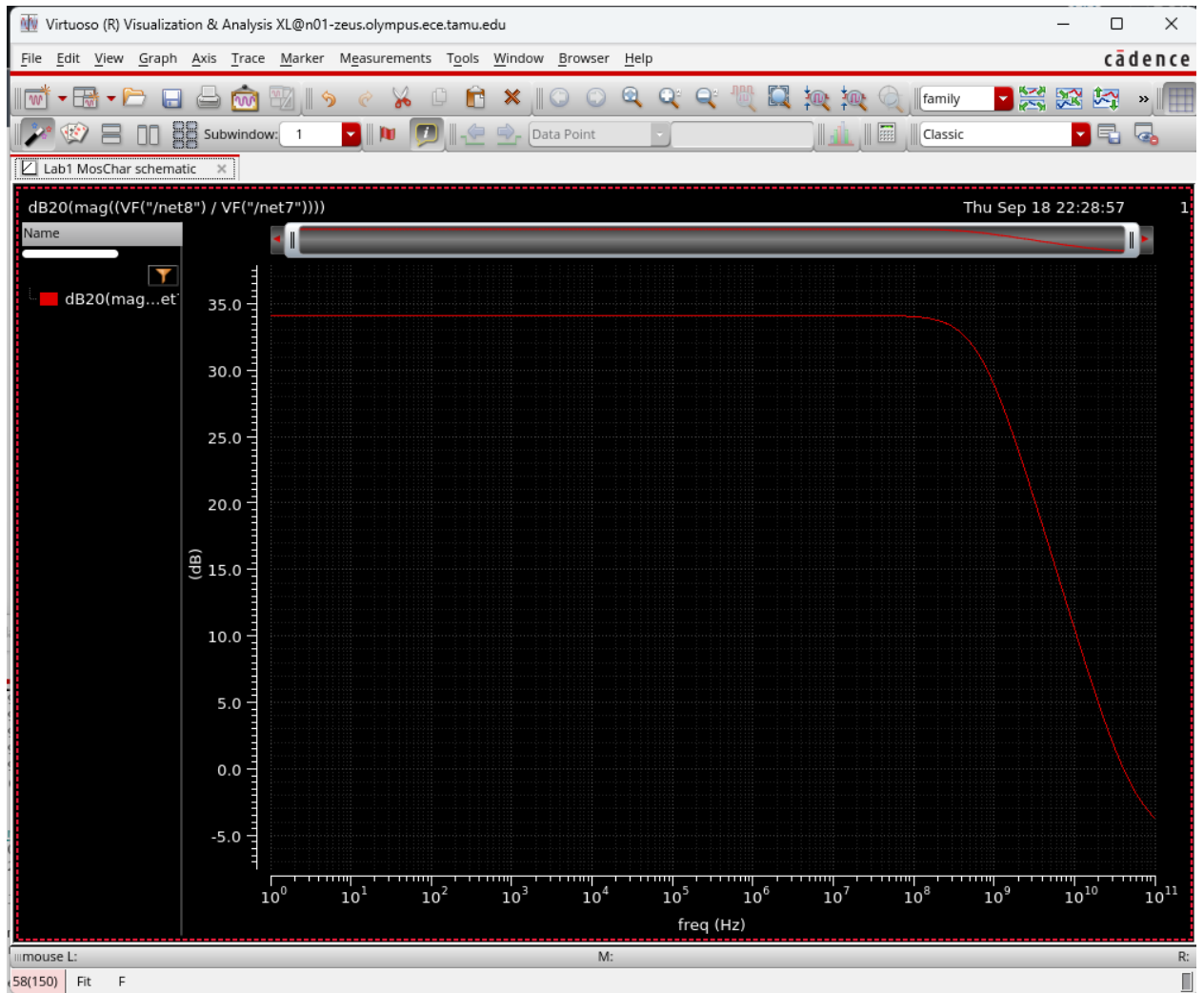


b. $V_{GS} = V_{tn} + 1/(40)$. Sweep V_{DS} from 0 to 1.8V, plot $(dI_D/dV_{DS})/I_D$ vs. V_{DS} and extract λ at $V_{DS}=1V$. Additionally, find the unity-gain frequency (f_t) using AC simulations.





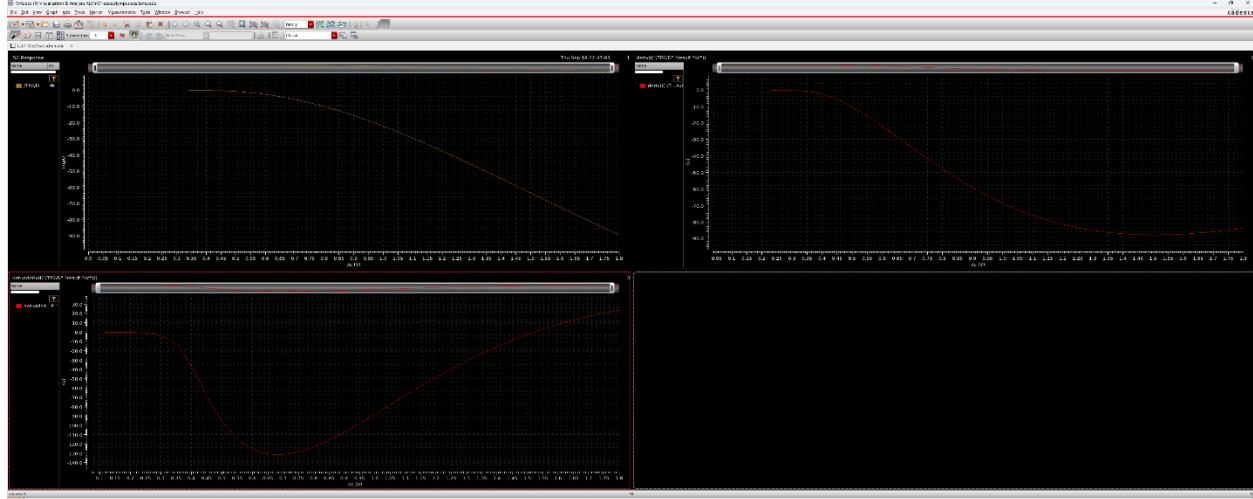
c. Reconfigure the circuit if necessary and find the intrinsic gain (A_i) using AC simulations when $V_{DS}=1V$ and $V_{GS}=V_{tn}+1/(4\theta)$. Refer to Appendix A for a method of checking the DC operating point before performing the AC simulations



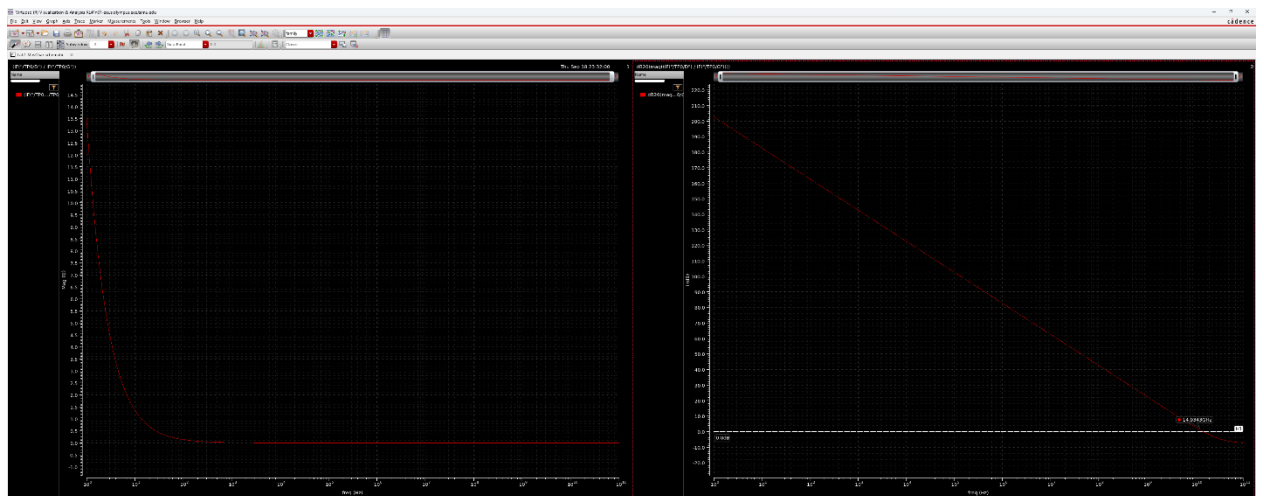
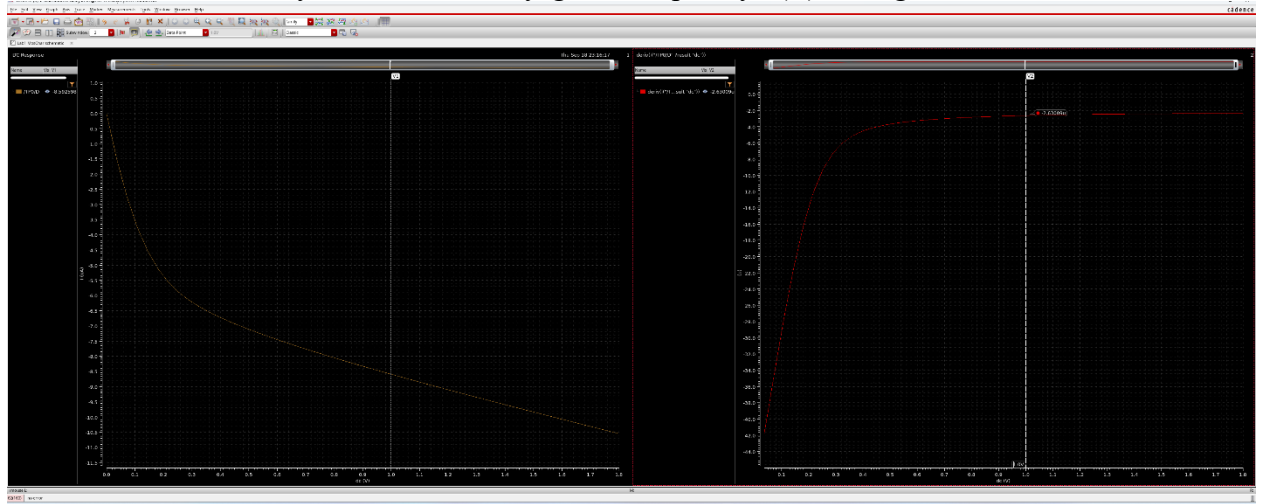
2. PMOS(pfetx):

5. $W/L = 0.5\mu/0.18\mu$ using 1 finger

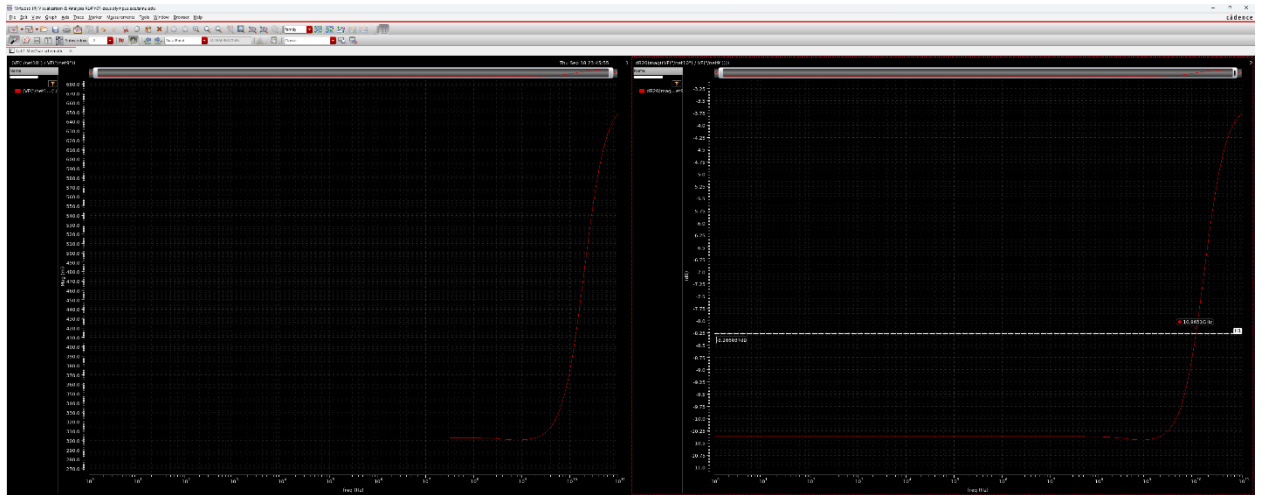
a. $V_{SD} = 1V$. Sweep V_{SG} from 0 to 1.8V, plot dI_D/dV_{SG} vs. V_{SG} , and extract V_{tp} , $\mu_p C_{ox}$ and θ



b $V_{SG} = |V_{tp}| + 1/(40)$. Sweep V_{SD} from 0 to 1.8V, plot $(dI_D/dV_{SD})/I_D$ vs. V_{SD} and extract λ at $V_{SD}=1V$. Additionally, find the unity-gain frequency (f_t) using AC simulations.

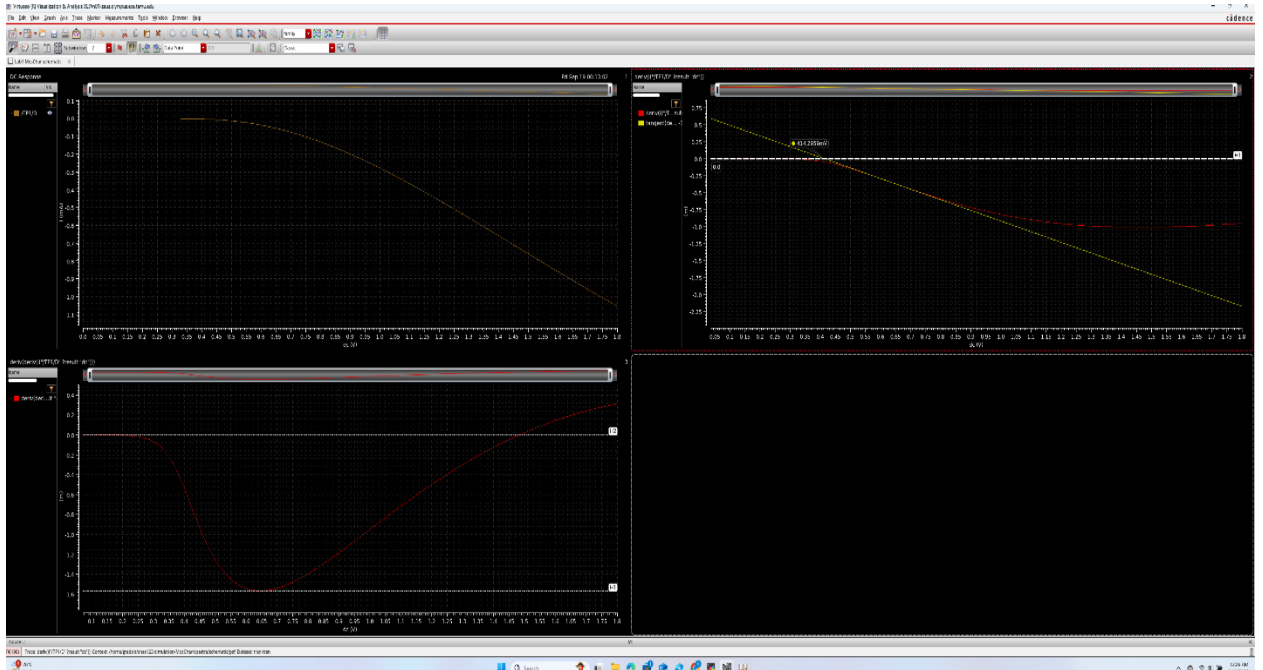


c. Reconfigure the circuit if necessary and find the intrinsic gain (A_i) using AC simulations when $V_{SD}=1V$ and $V_{SG}=|V_{tp}| + 1/(40)$. Refer to Appendix A for a method of checking the DC operating point before performing the AC simulations



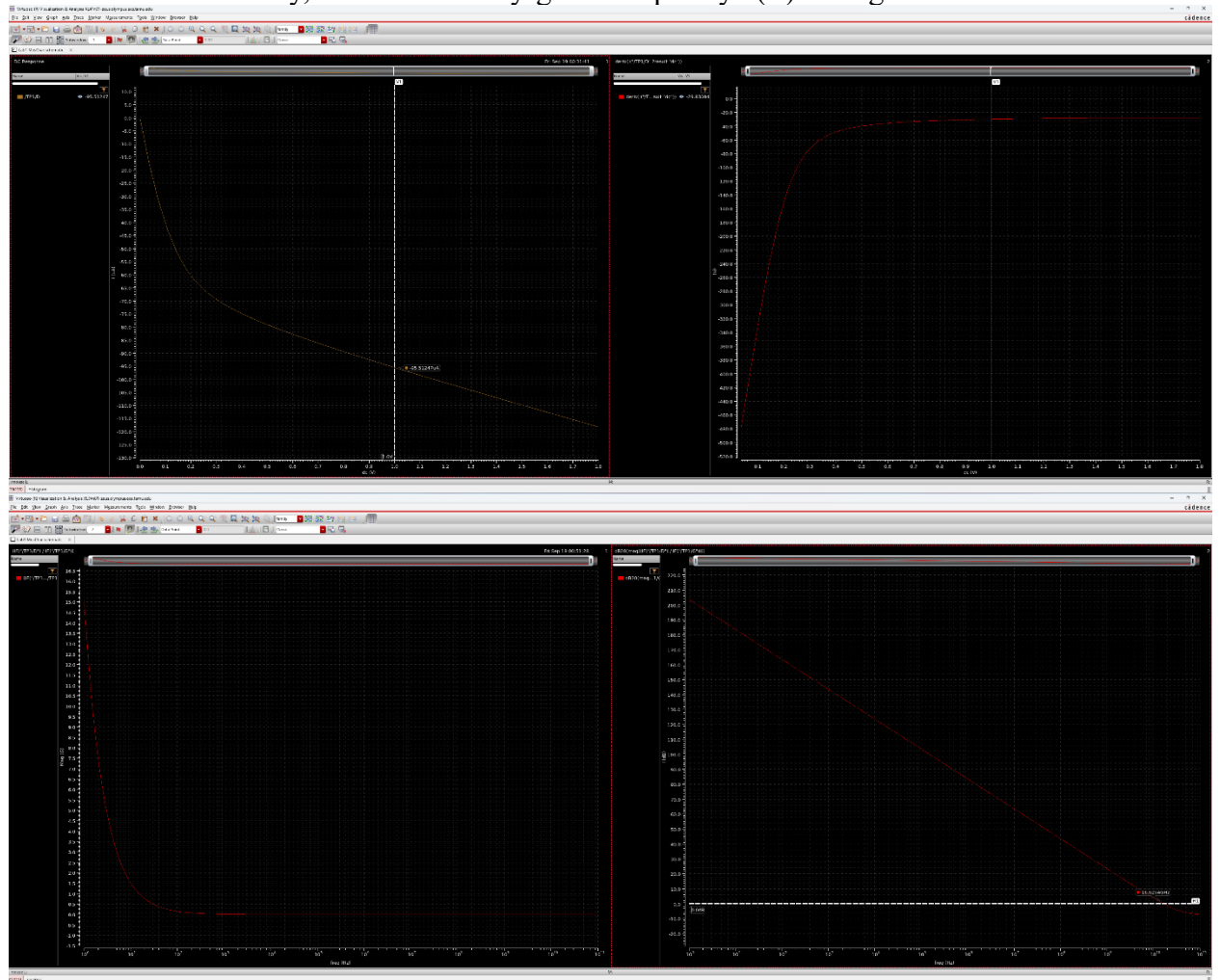
6. $W/L=5\mu/0.18\mu$ using 1 finger.

a. $V_{SD}=1V$. Sweep V_{SG} from 0 to 1.8V, plot dI_D/dV_{SG} vs. V_{SG} , and extract V_{tp} , $\mu_p C_{ox}$ and θ

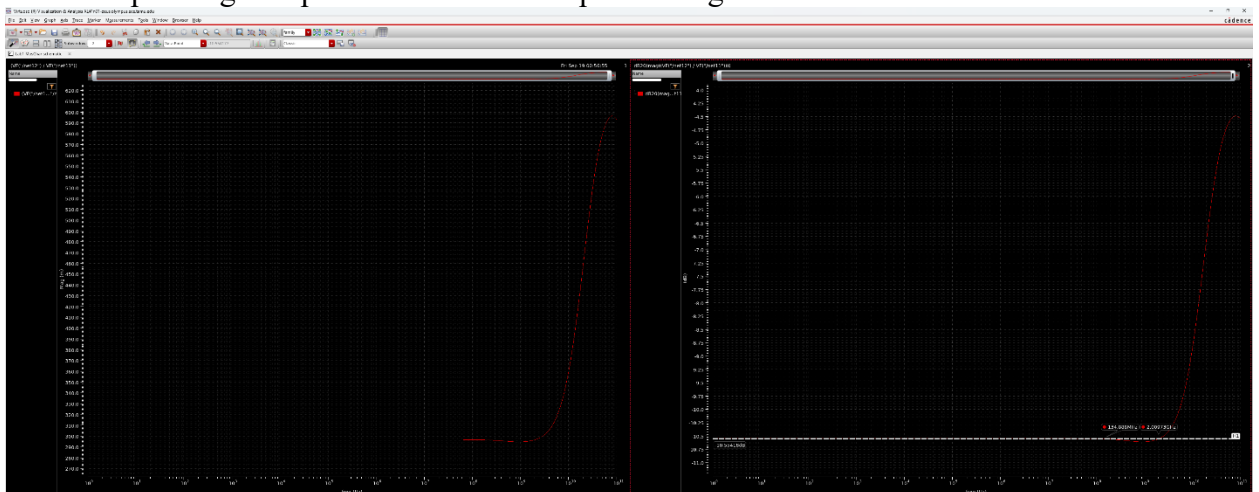


b. $V_{SG}=|V_{tp}|+1/(4\theta)$. Sweep V_{SD} from 0 to 1.8V, plot $(dI_D/dV_{SD})/I_D$ vs. V_{SD} and extract λ at

VSD=1V. Additionally, find the unity-gain frequency (ft) using AC simulations

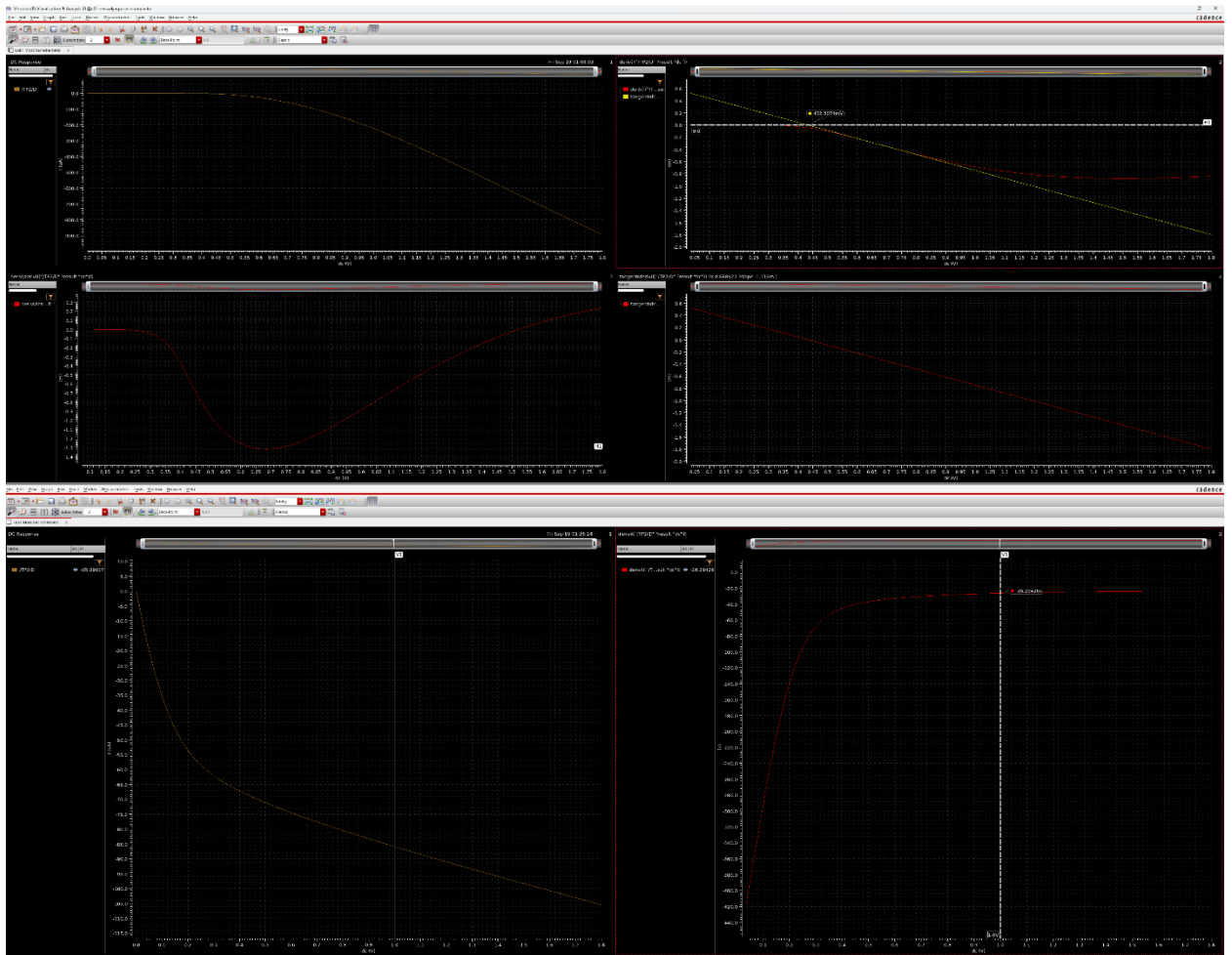


c. Reconfigure the circuit if necessary and find the intrinsic gain (A_i) using AC simulations when $V_{SD}=1V$ and $V_{SG}=|V_{tp}|+1/(40)$. Refer to Appendix A for a method of checking the DC operating point before performing the AC simulations.



7. W/L $5\mu/0.18\mu$ using 10 fingers

- a. $V_{SD}=1V$. Sweep V_{SG} from 0 to 1.8V, plot dI_D/dV_{SG} vs. V_{SG} , and extract V_{tp} , $\mu pCox$ and θ .



- b. $V_{SG}=|V_{tp}|+1/(40)$. Sweep V_{SD} from 0 to 1.8V, plot $(dI_D/dV_{SD})/I_D$ vs. V_{SD} and extract λ at

VSD=1V. Additionally, find the unity-gain frequency (ft) using AC simulations.

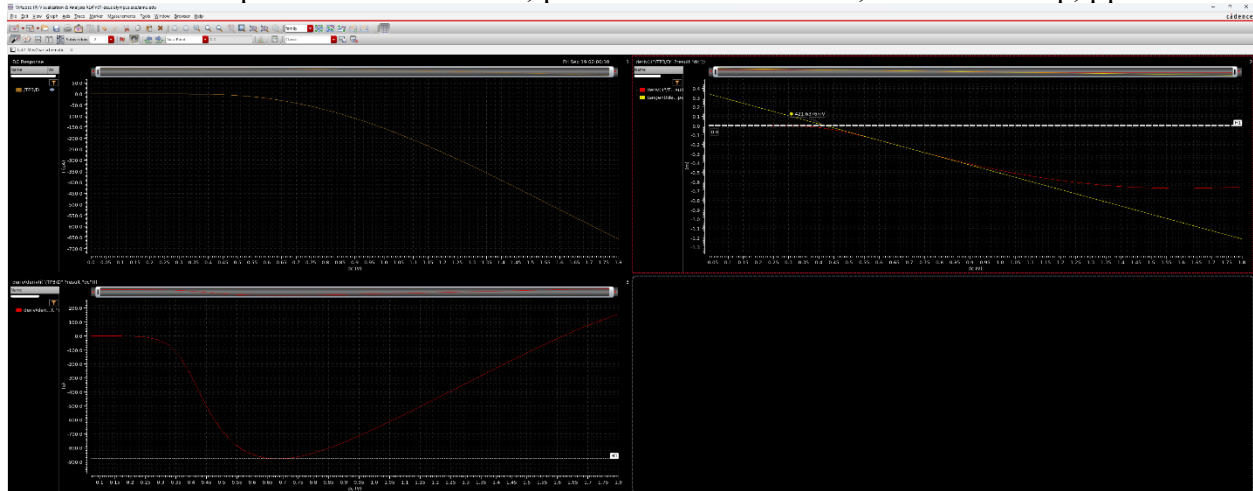


c. Reconfigure the circuit if necessary and find the intrinsic gain (A_i) using AC simulations when $V_{SD}=1V$ and $V_{SG}=|V_{tp}|+1/(40)$. Refer to Appendix A for a method of checking the DC operating point before performing the AC simulations

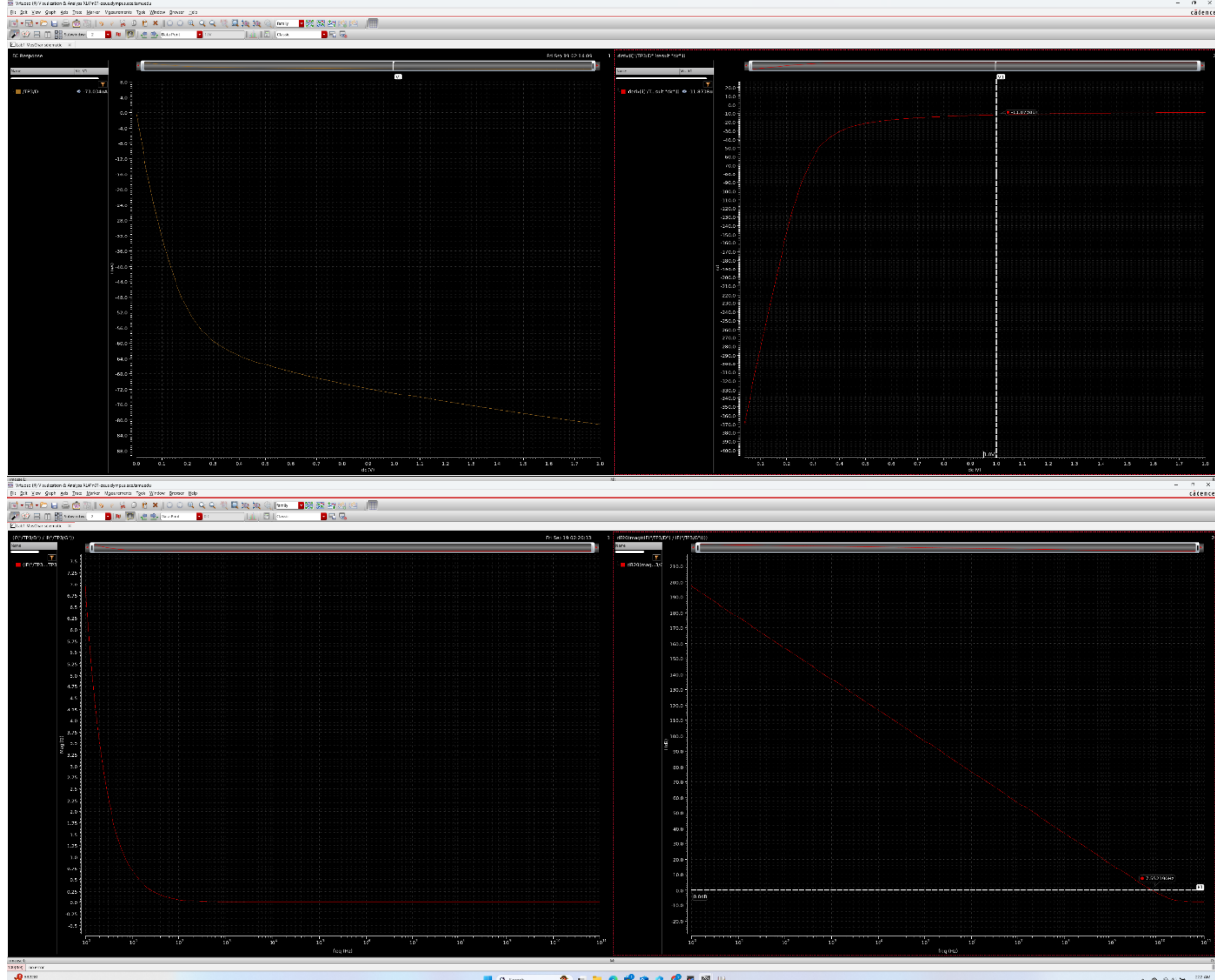


8. $W/L=5\mu/0.27\mu$ using 1 finger.

a. $V_{SD}=1V$. Sweep V_{SG} from 0 to 1.8V, plot dI_D/dV_{SG} vs. V_{SG} , and extract V_{tp} , μC_{ox} and θ



b. $V_{SG}=|V_{tp}|+1/(40)$. Sweep V_{SD} from 0 to 1.8V, plot $(dI_D/dV_{SD})/I_D$ vs. V_{SD} and extract λ at $V_{SD}=1V$. Additionally, find the unity-gain frequency (f_t) using AC simulations



C. Reconfigure the circuit if necessary and find the intrinsic gain (A_i) using AC simulations when $V_{SD}=1V$ and $V_{SG}=|V_{tp}|+1/(40)$. Refer to Appendix A for a method of checking the DC operating point before performing the AC simulations

